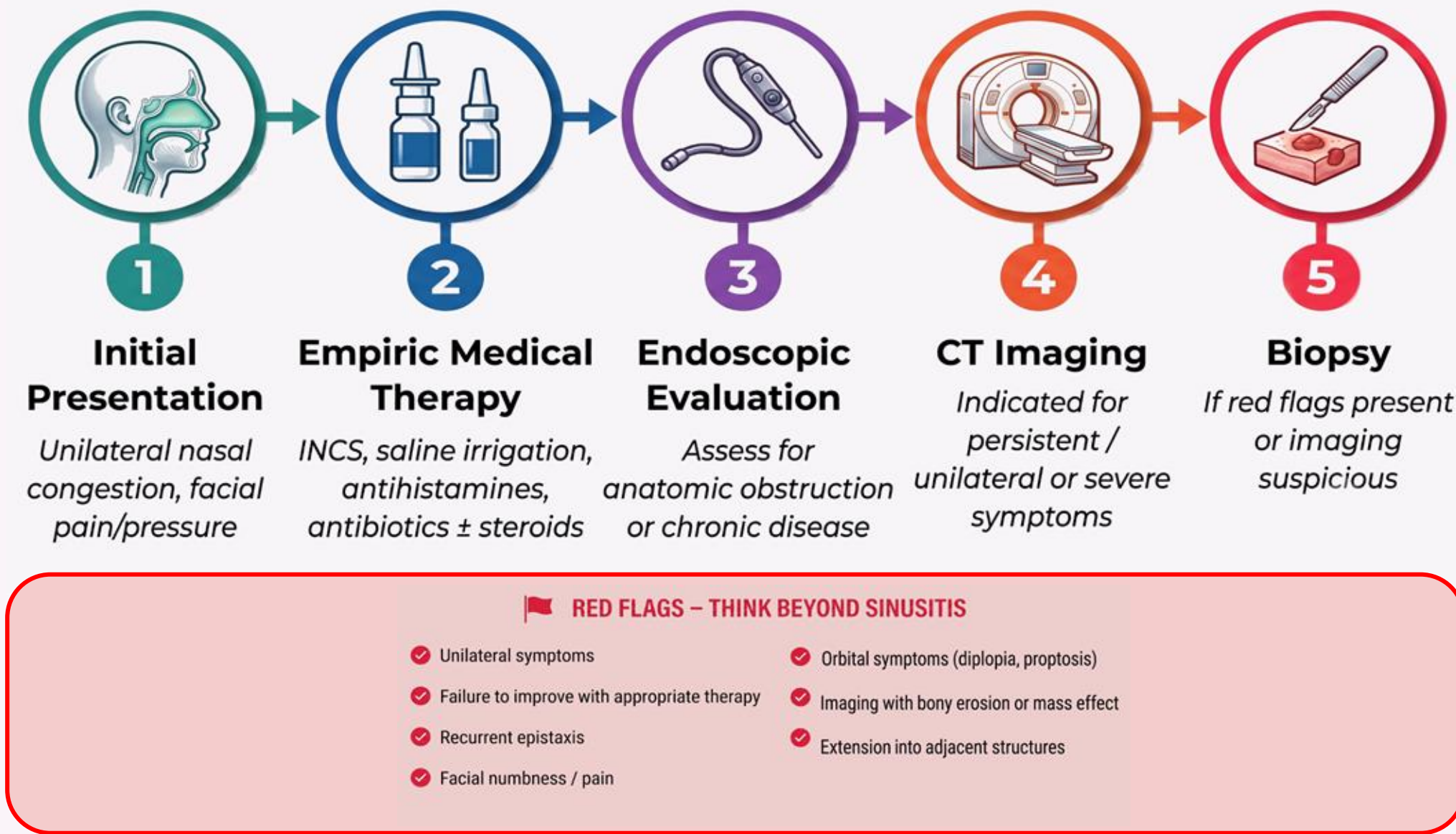


A Refractory 'Sinus Infection' with an Unexpected Diagnosis

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DIAGNOSTIC APPROACH TO REFRACTORY UNILATERAL SINONASAL SYMPTOMS



CASE AT A GLANCE

PATIENT & PAST MEDICAL HISTORY TRACK



PATIENT
68-year-old female



PAST MEDICAL HISTORY

- Hypertension
- Hyperlipidemia
- Graves' disease s/p radioactive iodine therapy
- Recently diagnosed breast cancer (undergoing radiation therapy)

DIAGNOSIS TRACK



PRESENTATION

Persistent right-sided nasal congestion, facial pain and pressure refractory to multiple courses of antibiotics and corticosteroids



CT FINDINGS (1/27/26)

Right maxillary sinus mass-like opacification with bony erosion and extension into adjacent structures



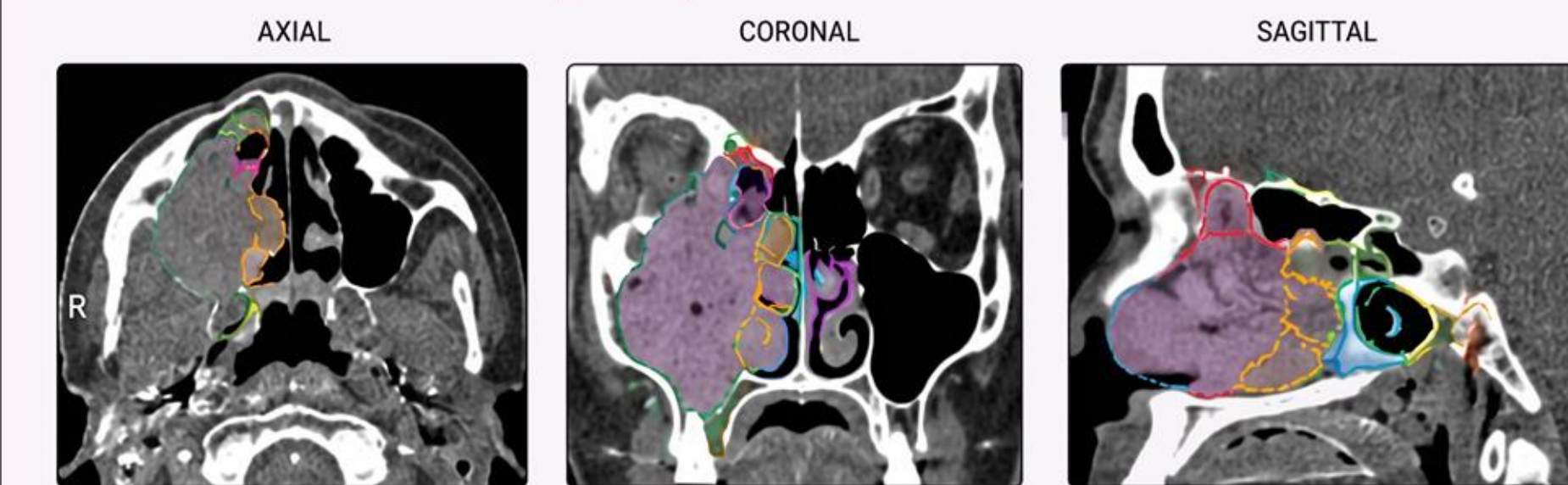
DIAGNOSIS
Diffuse Large B-Cell Lymphoma (DLBCL) of the Sinonasal Tract

TREATMENT PLAN

Referred to Medical Oncology for staging and systemic chemoimmunotherapy

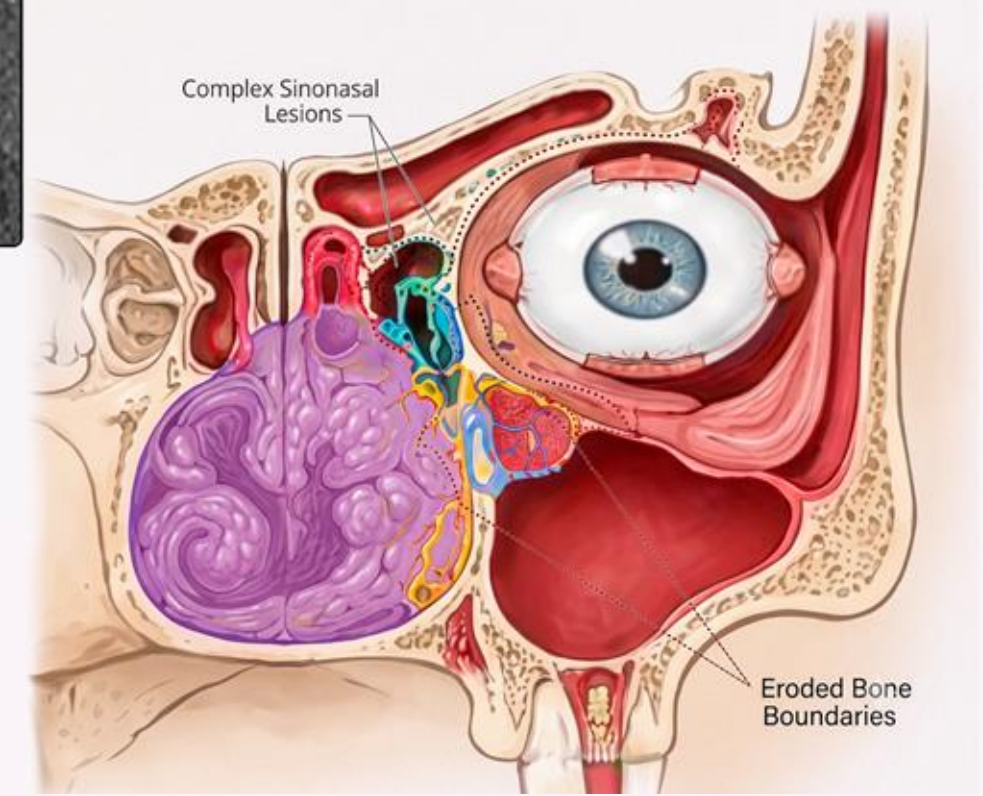
IMAGING FINDINGS

CT SINUS WITHOUT CONTRAST (1/27/26)



- Opacification / Mass-like expansion of right maxillary sinus
- Erosion into right nasal cavity
- Extension to middle ethmoid air cells
- Expansion of orbital floor with upward bowing
- Extension to pterygopalatine fossa
- Areas of bony erosion

- KEY FINDINGS**
- Complete opacification of right maxillary sinus with mass-like expansion
 - Erosion of right orbital floor and lamina papyracea with displacement of inferior & medial rectus muscles
 - Extension into right nasal passage, middle ethmoid air cells, and pterygopalatine fossa



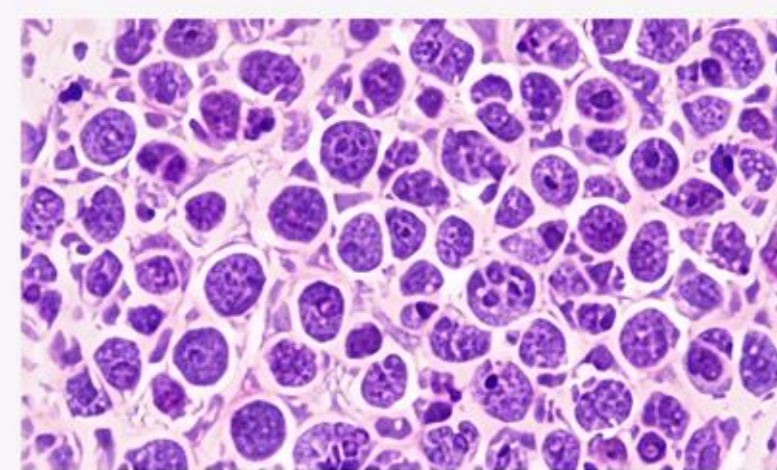
NASAL ENDOSCOPY



Large obstructive mass occupying the right nasal cavity

HISTOPATHOLOGY

Diffuse sheets of large atypical lymphoid cells.



Immunophenotype

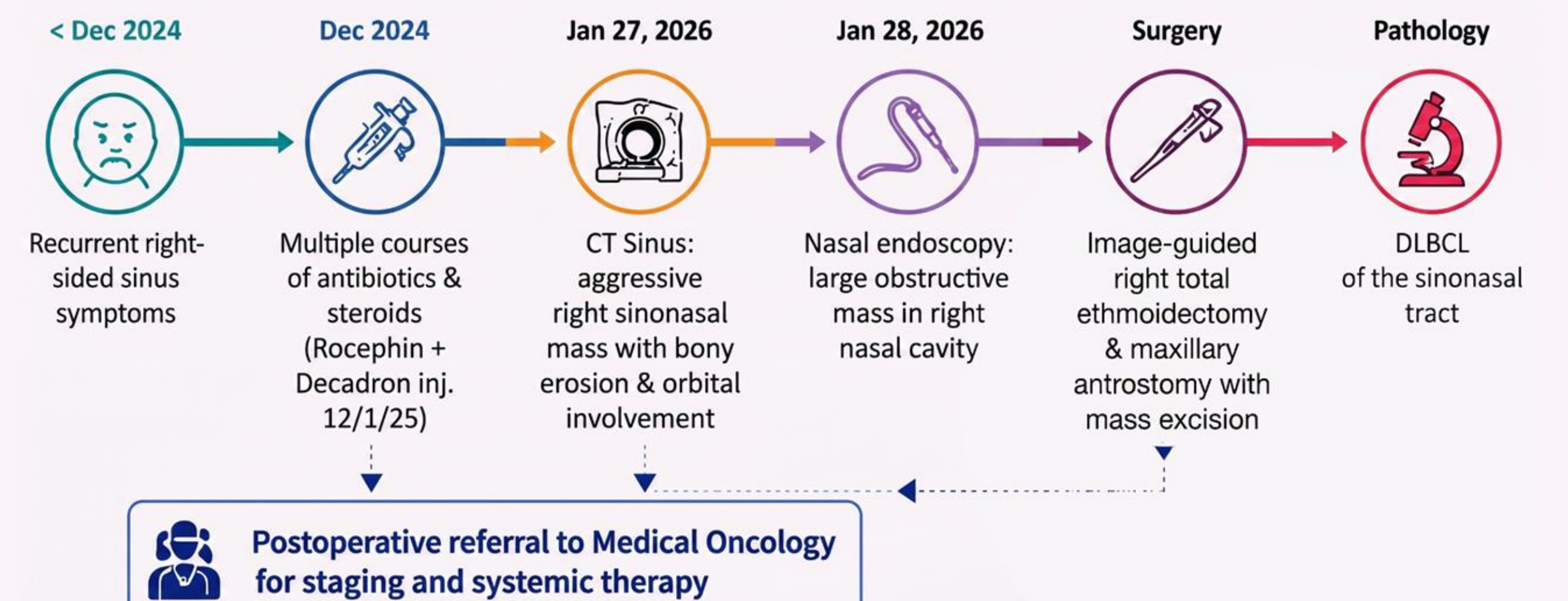
CD20+ BCL6+ MUM1+ Ki-67 high

Consistent with **Diffuse Large B-Cell Lymphoma (DLBCL)**

DIFFERENTIAL DIAGNOSIS OF AGGRESSIVE SINONASAL MASSES

Entity	Key Features	Distinguishing Clues
Inverted Papilloma	Benign but locally aggressive Schneiderian tumor	CT: remodeling > destruction Endoscopy: cerebiform pattern
Sinonasal Carcinoma	Epithelial malignancy (squamous, adenoid cystic, etc.)	Destructive bone changes Perineural spread
Fungal Infection (eg. Mucormycosis)	Angioinvasive fungal disease Immunocompromised hosts	Rapid progression Necrosis, black eschar
Inflammatory Disease (eg. GPA)	Granulomatous inflammation Systemic features	ANCA+, multi-organ involvement
Lymphoma (DLBCL)	Extranodal non-Hodgkin lymphoma Rare in sinonasal tract	Submucosal mass, bone erosion Histopathology confirms

PATIENT JOURNEY TIMELINE



KEY LEARNING POINTS

- Refractory unilateral sinus symptoms warrant evaluation beyond chronic sinusitis.
- Red flag features and destructive imaging findings should prompt tissue diagnosis.
- Sinonasal lymphoma, though rare, is a critical consideration in the differential.
- Early recognition and diagnosis of DLBCL allows timely initiation of highly effective therapy and improves outcomes.



NOT ALL SINUSITIS IS BENIGN.
THINK BROAD, IMAGE EARLY,
BIOPSY WHEN IN DOUBT.
EARLY DIAGNOSIS CAN BE
LIFE CHANGING.

TAKE-HOME MESSAGES



Persistent or progressive unilateral sinonasal disease with concerning imaging findings should always raise suspicion for an underlying malignancy.

biopsy. diagnose. treat.

ANATOMICAL EXTENT OF DISEASE (CT CORONAL SCHEMATIC)

- Right maxillary sinus (complete opacification)
- Right nasal cavity (erosion/extension)
- Middle ethmoid air cells
- Pterygopalatine fossa
- Orbital floor (erosion)
- Displacement of extraocular muscles

